Course on Introduction to Demographic Methods

Institute of Sociology and Demography (ISD), University of Rostock, Germany

Instructor Info

Aliakbar Akbaritabar, PhD

Office Hrs: by email agreement

ISD, 034 (phone: 4047); MPIDR, 215 (phone: 238)

https://akbaritabar.github.io

aliakbar.akbaritabar@uni-rostock.de

Course Info

Pre-requirement: None

Thursdays

9:00-11:00

PC pool 227, ISD

Lab Info

Every other week

9:00-11:00

PC pool 227, ISD

Overview

In this course, we will study fundamental concepts, measures and models that demographers use to understand population dynamics. Demography as a discipline relies on a core methodological component that addresses problems related to:

- Measuring and estimating population rates;
- Developing models to expand our analytical thinking;
- Formalizing theories;
- Linking individual-level processes with macro-demographic trends;
- Producing forecasts that combine all the available information.

Learning and qualification objectives

By the end of this course, students will obtain skills and capabilities in some or all of the areas listed below.

- Estimation of basic demographic parameters.
- Capability to meaningfully interpret demographic paramters.
- Ability to criticially evaluate demographic data and results that were published in the media.
- Ability to communicate demographic developments with scientists as well as with the the general public.

Reading Materials

Every week's materials include required and recommended readings specific to that week's topic, and some of the main ones are listed below.

Required Text

Wachter, K. W. (2014, June). *Essential Demographic Methods*. Harvard University Press.

Recommended Texts

Alexander, M. (2026). *Advanced Data Analaysis course in Demographic Methods*. Retrieved April 8, 2026, from https://github.com/MJAlexander/soc6708

Merli, M. G., Moody, J., Verdery, A., & Yacoub, M. (2023). Demography's Changing Intellectual Landscape: A Bibliometric Analysis of the Leading Anglophone Journals, 1950–2020. *Demography*, 10714127. https://doi.org/10.1215/00703370-10714127

Preston, S., Heuveline, P., & Guillot, M. (2000). *Demography: Measuring and Modeling Population Processes*. Wiley.

Acknowledgement on use of previous lecture materials

During this course, I will be introducing and using some materials from previous lectures on introduction to demographic methods or closely related and relevant topics. Here are some of them. I sincerely thank them for sharing their materials with me or providing them publicly:

- Emilio Zagheni's 2018 course
- Monica Alexander's 2025-2026 course (Alexander, 2025, December 5/2026): https://github.com/MJAlexander/soc6708
- Alyson van Raalte's 2024 EDSD course
- University of California (UC) Berkeley's 2024 formal demography workshop (this year's additional focus was on migration)
- Ernesto Amaral's 2018 lecture
- LSHTM course on Introduction to Demographic Methods and Population Analysis https://open.lshtm.ac.uk/mod/resource/view.php?id=7416
- Applied Demography Toolbox: https://applieddemogtoolbox.github.io
- R packages that provide analysis tools/functions (Abel, 2025; Abel et al., 2024; Hyndman, Booth, et al., 2025; Hyndman, Tang, et al., 2025; Raalte, 2019, November 27/2025; Riffe, 2024, 2025; Riffe et al., 2025; Yeung et al., 2022)

FAQs

How to access the course materials?

There is a private GitHub repository where all students are added in the first session. Every week's slides, hands-on materials, code, and data will be uploaded there.

What to do at home before/after sessions?

1) Study the reading materials and prepare some questions or discussion points. 2) Run codes in hands-on materials, reproduce the examples, extend them, and note the unclear points.

Do I need programming skills to attend? Where to learn the basics of R, Python, SQL?

No background knowledge is required, but some familiarity will help you advance faster. The course materials cover the basics and more advanced topics. Start by practicing with the provided examples. Materials also provide links to other introductory courses.

Can we use Artificial Intelligence to help us?

AI is a tool similar to any other tool out there. You should be "in control", and take "responsibility" for using it. Follow the ISD institute's reporting protocol to share how you used it.

Where to learn about academic writing?

1) https://writingcenter.gmu.edu/writing-resources/imrad/writing-an-imrad-report
2) https://www.nature.com/scitable/topicpage/effective-writing-13815989/

Grading Scheme

The final grade will be on a 1-6 scale with these meanings: 1.0: "sehr gut"/ Excellent; 1.3: "sehr gut"/ Excellent; 1.7: "gut"/ Good; 2.0: "gut"/ Good; 2.3: "gut"/ Good; 2.7: "befriedigend"/ Satisfactory; 3.0: "befriedigend"/ Satisfactory; 3.3: "befriedigend"/ Satisfactory; 3.7: "ausreichend"/ Pass; 4.0: "ausreichend"/ Pass.

Active participation in course discussions during the semester will inform my evaluation of your understanding of the topics. If you have specific concerns that might inhibit your active participation in class discussions of the reading materials, please inform me at the start of the course.

Course Examination

The mode of examination is a "term paper" ("Hausarbeit" in German). In the penultimate week, I will share a list of topics (from the course schedule below) and ask you to select one. You will use the reading list for that topic, search for other literature, plus use the data sources and discussed research questions, and write an essay that should include these sections: a) introduction, b) background and literature, c) methods and analytical strategy, d) results, e) discussion and conclusion (including possible limitations of your method and results), f) references (to the literature, data sources, etc., that you used).

Diversity and Inclusivity Statement

To become a great social scientist, a student needs to learn to add a "why" before every statement. The same statements that others hear without questioning. Students need to develop sharp observation skills and find a new angle in seeing familiar social phenomena. This is how they can observe inequalities and previously less explored aspects of society. This is the same society in which we all live, however, those with sharpened observation skills can identify inequalities in a familiar context. They can question the ordinary, find reasons behind observed phenomena, and imagine counterfactual scenarios for the future. This practice helps students develop critical thinking skills. I converse with students, write their ideas on the whiteboard, and ask them to explain those ideas. I create a safe space where every student feels welcome to contribute, which helps students develop their own ideas confidently.

Students with Disabilities

Please inform me as soon as possible of disabilities or conditions that you may have. The sooner you notify me, the better I will be able to coordinate with the institute and other responsible persons to provide the best possible support.

Students' Mental Health and Well-being

Academic life and university studies can be very stressful and challenging. Please inform me or seek proper professional help if you are concerned with your mental health and well-being. Seeking help is one of the most courageous things to do.

Integrity, Plagiarism, and Ethics of Research

Students are expected to prepare their final assignments independently. Any violation of this will not be tolerated and will be appropriately reported to the faculty and university according to the institution's guidelines. If you use a resource in your work, cite it properly and give due credit by following established guidelines, e.g., APA: https://apastyle.apa.org/style-grammar-guidelines/citations/plagiarism.

Credits

- The syllabus LaTeX template is modified based on https://www.overleaf.com/latex/templates/inzane-syllabus-template/zpgzsdfxvqmz.

Class and Lab Schedule

Topics covered in one or multiple weeks including additional hands-on lab sessions

1	Introduction	Why study Demography?
	Accessing course materials	Importance of reproducibility; Version control, Git, GitHub
2	Mortality	
3	Fertility	
4	Population projections	
5	Migration	
6	Kinship	
7	The future	Digital and Computational Demography
8	?	
9	Wrap up	Description of examination
10	Final examination	15 min presentation to share your selected topic + Q&A; 6 weeks to prepare a short essay and empirical analysis